

HOMEWORK 9 FOR MATH 2860 (ELEMENTARY DIFFERENTIAL EQUATIONS)

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Due at 10:00 EST in class

1. Define the **Laplace transform** of a function $f(t)$, defined for $t \geq 0$.
 - (a) Using the definition, compute $\mathcal{L}\{t * e^{-t}\}$
 - (b) Find the Laplace transform of the piecewise function

$$f(t) = \begin{cases} 0, & 0 \leq t < 3, \\ 5, & t \geq 3. \end{cases}$$

- (c) Find $\mathcal{L}^{-1}\left\{\frac{s+1}{s^2+4s+5}\right\}$.

2. Solve the IVP

$$y'' + y = \delta(t - \pi), \quad y(0) = 0, \quad y'(0) = 0.$$

3. Solve the IVP:

$$y'' + y = \sin x, \quad y(0) = 0, \quad y'(0) = 1.$$

Show all your work.